The Essence of Civilization

Who gives, who gains, who cannot leave—
and why every great order eventually falls

MACHARIA BARII

October 2026

Civilization is a ledger of human relations.

— Macharia Barii

The Question Beneath the Monuments

Every schoolchild knows what a civilization is. It is the pyramids. The aqueducts. The codes of law, the temples, the great canals. We count a people civilised by what they built, and we rank the ages by what they left behind.

Here is an awkward fact. Every one of those monuments was built by people who were, for the most part, not allowed to enjoy them. The pyramids were raised by corvée labour—peasants compelled by the state for part of each year. Roman law was brilliant, and it exempted Roman elites from most of what it prescribed for everyone else. Athenian philosophy was born in a city that ran on chattel slavery, and the geometry that measured the Earth was put to work dividing land among masters.

So the monuments prove something. They do not prove justice.

This essay is a plain-language version of a formal theory—a theory built out of game theory, dynamics, and careful definitions—whose central claim is simple enough to state in one line: **civilization is not what a society produces. It is the ledger of what people give to and take from one another.** The pyramids are the output. The ledger is the thing itself.

Once you see it that way, a great deal becomes clearer: why empires rise and fall on schedules that look almost mechanical, why the most creative ages are often the most cruel, why some orders can be reformed and others only break, why wars happen, and why the modern world

feels stable and unjust at the same time.

The Ledger

Imagine two people who must work together to survive—two farmers sharing an ox, two merchants pooling a ship's cargo, a ruler and the people who till the land. Between them runs an invisible ledger. On one side: what each puts in. On the other: what each takes out.

The theory asks three questions of every relationship, whether it is a marriage, a business partnership, an empire, or a trade deal:

1. **What does each party give?** Time, labour, goods, loyalty, risk.
2. **What does each party get?** Food, profit, protection, status.
3. **Can either party walk away, and can either party complain?**

That third question turns out to be everything.

Three Kinds of Relationships

Once you measure the flows, every human relationship falls into one of three types. The names come from biology, but the meaning is entirely about money, labour, and power.

Type	What happens	Historical example
Symbiosis	Both sides get enough to keep going, and both would rather stay than leave	Partners in a healthy firm; allies in a functioning alliance
Parasitism	One side is squeezed below what it would choose, but above what it needs to survive	Peasants under corvée; provinces under tribute; debt peons
Predation	One side is squeezed below survival itself; the host is being consumed	Slavery; conquest; a fishery stripped to collapse

The middle category is the important one, because it is where most of recorded history lives. A parasitic order can last a very long time. It can be legal, religiously blessed, and thoroughly admired by the people at the top. The peasants hate it, but they cannot leave—the land is elsewhere, the borders are closed, the debt is inherited—and so they endure. The ledger is unbalanced, but not yet fatally.

> The theory is not moralising. It says nothing about whether a relationship *ought* to be symbiotic. It says something harder and more useful: **unbalanced ledgers are unstable**, and it can tell you exactly how unstable, and why.

The Two Thresholds

Every person, and every group, has two thresholds.

The first is the **participation threshold**: what you would need to be offered to prefer this arrangement to your next-best alternative. If what you receive falls below it, you are being exploited—you would leave if you could.

The second is the **reproduction threshold**: what you need simply to keep existing, to stay healthy enough, fed enough, organised enough to keep working. If what you receive falls below *that*, you are being consumed.

These two thresholds define the battleground of history. Exploitation lives in the gap between them:

- Above both: symbiosis. Fair-enough.
- Between them: parasitism. You are exploited, but you survive—so the order survives.
- Below the second: predation. The host dies, and the party that was feeding on it dies with it.

The exploiters' eternal problem is that they cannot push as hard as they would like. Push a worker below survival and the worker collapses; push a province into genuine ruin and the province stops producing. The theory shows exactly how far extraction can go, and what limits it.

Two Things That Limit the Powerful

Only two forces cap extraction, and both are ancient.

Exit is your ability to walk away. Serfs could not—they were bound to the land. Slaves could not—they were property. Modern capital can: a wire transfer moves a billion dollars overnight, and a corporation can relocate its headquarters before lunch. Exit is an economic fact, not a virtue.

Voice is your ability to push back: to strike, to litigate, to vote, to protest, to riot. Voice is also an economic fact. It is cheap for small, tightly organised groups—a merchants' guild, a barons' revolt, a general's army—and ruinously expensive for large, scattered ones. A million peasants have more raw power than a hundred bankers, and almost none of the organisation. Throughout history, the heaviest taxes fall on those who cannot leave and cannot combine.

> **Why peasants were taxed and guilds were not.** A guild was small, rich, and could coordinate in a single room. A peasantry was vast, poor, and could not coordinate at all—each family that withheld labour risked starving alone while the others gave in. The tax collector knew this. So the ledger tilted, century after century, toward those who could combine and away from those who could not. The labour movements of the last two centuries were, in this language, one long attempt to make a large group cheap to coordinate.

Rulers imagine that the trick is to divide the many. The theory says the trick is subtler. Merely cutting a large class into smaller ones does not reduce its total power, because each piece can still organize—often more cheaply than the whole could. What actually weakens a class is raising the cost of combining at all: setting its members against one another by identity, spying on their meetings, and flooding the air with stories that keep them from agreeing on anything. Divide and rule is not the division. It is the engineering of distrust.

> The powerful do not fear a big group. They fear a **coordinated** one. That is why the technologies that let people organize—assembly, printing, the telegraph, the network—have always been treated as dangerous, and why their suppression is a policy and not an accident.

Why the Grandest Things Prove Nothing

Back to the pyramids. If civilization is the ledger, then its achievements are the *output* of that ledger, not the ledger itself. Achievements need surplus—food above subsistence—plus patience, organisation, and skill. All of those come from the ledger. But none of them requires fairness.

A patient, ruthless order can produce more monuments than a kind and chaotic one. The theory puts it sharply: **creativity and exploitation are not opposites; they are joint products of the same thing—time**. The same long horizon that lets a pharaoh plan a pyramid for thirty years lets him plan a tax system that will bleed the countryside for thirty generations. The Roman peace and the Roman aqueducts come from the same patience that made Roman provincial extraction so efficient.

> So achievements are not evidence of justice. They are **diagnostic**. A monument tells you who controlled the surplus, how long they expected to rule, and who was forbidden to complain. The record of what was built is really a fossilised record of the ledger.

The method that follows is three words long: **measure the ledger first**. Before admiring a civilisation's art, ask who commanded the labour, who bore the cost, and who was left out of the story.

The Patient Win—for a While

Why did Rome last five hundred years while the Viking raids lasted an afternoon? Not, the theory says, because Rome was kinder. Rome was a predator with a long time horizon; the raiders were predators with a short one. The theory models both as extractors of a host—a province, a population—and finds that the amount you can safely take depends on your **discount rate**: how much you value next year against this year.

A patient extractor, who values the future heavily, keeps the host alive, builds roads, administers justice, invests in ports—because a healthy host produces for centuries. An impatient extractor loots and leaves. The difference between an empire and a raid is not morality. It is patience.

But—and this is the dark corollary—patience cuts both ways. The same long horizon that makes an order build also makes it **extract well**. The pharaoh's corvée and the pharaoh's pyramids share a discount rate. The most creative societies in history are frequently among the most exploitative, and the theory explains why this is not a coincidence but a structural necessity.

The Race for a Shared Host

Patience works only if you are alone. A single patient lord conserves his forest, because the forest is his and will still be his next year. But most hosts in history have been shared: the pasture grazed by every herder in the village, the fishery open to every boat, the hunting ground of many bands, the tax base claimed by several lords, and now the atmosphere claimed by everyone and owned by no one.

When many extract the same host, they do not consult. Each takes what seems best given what the others are taking—and the arithmetic of that race is brutal. Competition acts like impatience. A predator who would have conserved a river alone will draw it down if a dozen rivals are drawing it down beside him, because anything he leaves behind will be taken by someone else. The most patient man in the race still races.

> The commons is not destroyed by greed. It is destroyed by arithmetic. **Enough competitors make even patient predators behave like raiders**, because the future you would otherwise protect is no longer yours to protect.

This is the old tragedy of the commons, and the theory gives it its real cause. It is not a failure of character; it is a failure of numbers. A single extractor conserves; a crowd annihilates. The communities that survived their commons—the mountain pastures, the irrigation ditches, the fishing villages that governed themselves—did something specific: they created rules and enforcers that turned many predators into one. That is not a moral achievement. It is the construction of an institution.

The one who benefits from fewer rivals. Here is the fact that changes everything: every predator loses from the race. If a rival exits, each survivor takes more, and the host lasts longer. So the extractors themselves have a reason to want rivals gone—and the strongest of them has a reason to remove the others by force. The race does not end in ruin by itself. It ends when someone wins it.

The Center Invents Itself

The bilateral ledger had no center: two parties, one relation, no third actor. But the race among many predators generates a third actor on its own—the one who can exclude the others and become the single extractor of a host.

This is the origin of the state told as economics. A warlord who clears his region of rival warlords does two things at once: he takes a larger share, and he leaves the host alive longer than the free-for-all would have. The host benefits from monopolization. So do the remaining predators. Both pressures push the same way, which is why centers arise so reliably—and why their absence is so reliably followed by what the chronicles call dark ages.

> The state is not the opposite of predation. It is **predation organized**. Charles Tilly's sardonic line—that war made the state and the state made war—is, in this framework, a description of a cartel.

But the center is not free to charge whatever it likes. Its take is capped by what it actually provides. It keeps the host alive, which the race would not. It enforces agreements, so people can trade and cooperate across a distance. It keeps records and counts what it governs, so that cheating is harder to hide. A center that does these things can charge for them. A center that stops doing them is simply a warlord again, and will be replaced by one who does.

Then comes the question every empire faces from inside: who guards the guardian? The center is itself an extractor with its own horizon. If it expects to be overthrown soon—if succession is a knife-fight, if coups and revolutions are frequent—it will loot, because the future is not its to enjoy. If it expects to endure, it will build, because a province that is still rich in thirty years still pays taxes in thirty years. The difference between a dynasty that dug canals and one that emptied the treasury is not virtue. It is the probability of surviving the night.

> **Horizons, not virtues, build the works of civilizations.** When a ruler's future is insecure, he behaves like a raider no matter how refined his manners. The institutions that lengthen a ruler's shadow—succession rules, courts that outlive the king, contracts a ruler cannot revoke—are not decorations. They are what turns a predator into a builder.

The Chain of Extraction

No empire extracts in one step. It extracts in layers. A peasant pays a landlord, the landlord pays a provincial governor, the governor pays the capital. Value climbs a chain, and every link keeps a cut on the way up.

Two facts follow, and both are visible in every long-lived empire. First, the chain leaks. For every measure that reaches the top, the bottom keeps less the more layers stand between the field and the throne—not because any one layer is cruel, but because many modest cuts compound into a heavy one. The peasant is not taxed by the emperor. He is taxed by everyone between himself

and the emperor.

Second, and more surprising, the chain breaks not at the top but at its weakest link. Each layer must keep enough of its own people willing to stay; when a layer fails that test, it stops forwarding upward, and every layer above it starves while every layer below it is suddenly left alone. This is why revolts so often begin at the periphery of an empire—Judaea, Gaul, the Andes, the frontier—rather than in the capital. The capital may be rich and stable while the province that feeds it is beyond breaking. The center falls not because it is weak but because a link far from it gave way.

Every long chain of command is a long chain of leakage **and** a long chain of fragility. It fails where it is weakest, which is almost never where it is richest.

The chain you live in. The same shape appears wherever value is carried across distance and middlemen: tax farming in the ancient world, the manorial rent roll in the medieval one, the platform that sits between the user and the advertiser, the brand between the factory and the shopper. In each, most of the cost is borne at the bottom, most of the visible reward is captured above, and the whole thing is vulnerable at whichever link holds its people least securely.

The Price of Force

An order that lives by squeezing must also pay for the squeezing. Every act of extraction that people resent has to be enforced: with soldiers, with police, with courts, with clerks. That enforcement is expensive, and the expense rises with the injustice—the more you take beyond what people will accept, the more force you need.

But force is not the only way to keep people compliant. The cheaper way is **belief**: the feeling that the arrangement, however painful, is proper. Religion is the oldest such technology, but it is not the only one. Nationalism, the sacredness of law, faith in the market, trust in progress, the charisma of a leader—all of these perform the same function. They lower the price of compliance.

Belief does not add force to an order. **It subtracts the price of force.** That is why every durable civilisation invests so heavily in stories, rituals, holidays, and holy books—and why, when belief drains away, the treasury must suddenly pay for what the temple used to provide for free.

Religion is the most interesting case because of a strange feature: its central transaction is with a partner nobody can see. No one can measure what the gods contribute, or check what they promise. That unverifiability is precisely what makes religion so *cheap* as a legitimising technology—no audit can ever contradict it—and precisely what makes it so *dangerous* in the

hands of power. Whoever speaks for the unverifiable partner holds a monopoly on the excuse. The theory is careful here: it makes no claim about whether the gods exist. Its claim is narrower and harder to dismiss: **the ledger cannot audit heaven, so the audit must fall on the institution that claims to speak for it**. When a preacher, or a party, or a platform demands a sacrifice in the name of something no one can check, the scepticism belongs not on the heavens but on the messenger.

And belief, like everything else, can be stolen. A rival story—a reformation, a new ideology, a counter-narrative—does not have to defeat the army to bring down an order. It only has to drain the belief that made the army cheap.

There is a harder reason belief is scarce at scale. Belief is shared—everyone in the order benefits from a common story of why the order is proper—but the work of producing it falls on particular people: priests, teachers, poets, propagandists. Whoever pays that cost pays it for everyone, and everyone else gets the benefit for free. This is the free-rider problem turned on a society's self-understanding. A small, tight order can hold its story together. A large, diverse one cannot produce belief fast enough, because the people who would produce it have little reason to bear the cost alone.

That is why scale and surveillance travel together. When an order grows too large or too varied to keep producing belief voluntarily, it substitutes coercion for conviction: censors, informers, spies, a police that makes dissent expensive. Surveillance is not a sign of strength. It is what an order buys when it can no longer sell belief at scale.

Justice: The Rebalancing

So what keeps an order alive? Not goodness. **Rebalancing.**

Every durable civilisation contains machinery that pushes the ledger back toward balance when it drifts too far. The mechanisms are always the same four:

1. **Transfers**—debt cancellations, jubilees, land reforms, taxes returned as bread.
2. **Rights**—limits on what may be taken, written down and enforced.
3. **Procedures**—courts, votes, the right to organise: ways of raising voice and lowering exit costs.
4. **Repair**—restoring what predation consumed: the freed serf, the rebuilt farm, the cancelled fine.

The ancients knew all of this. Mesopotamian kings cancelled debts at accession. Solon wiped out Athenian debt bondage in a single act. Roman citizenship was extended, slowly and grudgingly, until almost everyone inside the empire held it. These were not kindnesses. They were the periodic rebalancing without which the ledger—and the order—would have broken.

> A civilisation is not judged by whether it contains injustice. All of them do. It is judged by whether its rebalancing machinery is **faster than the drift toward injustice**. When it is not, the debt simply grows until something breaks.

There is also a question of *how* a ledger gets balanced when it is far out of true. Sometimes it can be done in steps—each move small enough that no group, once it has moved, wants to walk back. When such a path exists, an order can reform itself, however slowly. When it does not—when every possible division of what the society produces leaves some group better off overturning everything—then no settlement is stable, and change can only arrive as a jump: a revolution, a secession, a jubilee. That is the formal difference between a society that can mend itself and one that can only break.

The difference often comes down to who has a seat. A settlement holds only if everyone it affects is party to it. When an order quietly drains people it never invited to the table—the province taxed without representation, the community polluted without consent, the generation that pays for a debt it never incurred—it manufactures the very grievance that makes a stable settlement impossible. Widening the table is not merely just. It is what makes reform possible instead of force.

Why Civilizations Fall

Civilisations do not usually die of conquest. They die of arithmetic.

Take an order that extracts, that believes its own justifications less every year, and that has no functioning rebalancing left. The theory proves that such an order must collapse in finite time. Squeeze, and grievance grows; grievance drains belief; drained belief means compliance must be bought with force; force costs money, which is raised by squeezing harder; and the squeeze shrinks the very capacity the order depends on. It is a circle, and it only spins one way.

What makes history look cyclical is that the reset is built in. When the debt, the grievance, and the empty treasury finally break the order, what follows is not nothing: it is a **jubilee**, a land reform, a new dynasty, a new constitution. The old ledger is torn up, shares are reset, belief is briefly abundant—and the cycle begins again. China's dynasties, Mesopotamia's debt cancellations, Rome's periodic extensions of citizenship: the same machine, running for three thousand years.

> Collapse and renewal are not opposites. They are the two halves of a **rebalancing that came too late**.

And then there is the oldest machine of all.

> War and conflict are the results of a failure in the ability to reason together and to negotiate on equal terms as equal human beings: the failure of the civilisation process itself.

The theory's account of war follows directly. Two parties with a shared ledger and a grievance can normally settle—the stakes are known, the parties are two, a price can be found. War begins when that becomes impossible: when there are too many parties, when no one can verify what a rival will do once the balance shifts, when the bargaining itself collapses into guessing. War

is what the ledger looks like when rebalancing has failed at scale. It is the most expensive rebalancing mechanism ever invented, and it is always, in the end, a confession.

There is a colder mechanism at work in alliances. A coalition formed against a common enemy carries the seed of its own consolidation. While the threat lasts, leaving the coalition is more dangerous than staying, so the strongest member can raise its demands without changing any formal equality—the smaller allies cannot afford to walk away with the enemy still outside. The Delian League became the Athenian empire this way, and the pattern repeats wherever a league outlives its fear. When the threat finally ends, the subsidy collapses: the leader must either share power or watch the coalition dissolve.

When One Failure Becomes Everyone's

Everything so far has treated a civilization as a single ledger with parts. But no civilization is one ledger. It is a web of them—provinces tied to a capital, banks to borrowers, farms to cities, suppliers to factories, each relation resting on the others. Once the parts are connected, failures can travel.

The logic is simple and merciless. A node in this web—a region, a firm, a bank—depends on its neighbors to keep its capacity above the level it needs to keep going. If a neighbor fails, it loses some of what sustained it, and can be pushed below its own threshold in turn, dragging its neighbors down with it. Whether such a failure stays local or consumes the whole web depends on how densely the web is connected and how fragile each link is. Below a certain density, a shock burns itself out; above it, a single failure can run through everything.

> In a network of ledgers, **the death of one node can be the death of its neighbors**. Above a certain density of connection, collapse stops being an event and becomes a contagion.

Not all nodes are equal. The failure of a well-connected point—a financial center, a breadbasket province, a canal, a single supplier everyone relies on—is far more dangerous than the failure of an isolated one, because everything it touches loses support at once. This cuts two ways. The cheapest way to topple a network is not to attack it everywhere but to attack what connects it. And the cheapest way to protect a network is to reinforce the same points. The most consequential extraction, in a connected order, is not broad. It is *targeted*.

And connectivity is double-edged. The integration that makes an order productive—more relations, more trade, more surplus—is the same integration that makes its collapse systemic rather than local. Rome's roads carried grain and legions; they also carried the failure of one region into the next. The collapse of a single ledger is a death. The collapse of a network is a contagion.

After the cascade. Rebuilding is slower than collapse, because what was destroyed was not only capital but trust—the confidence that agreements will be kept. When trust is low, coordination is expensive, and no one can organize anything large for a long time. What

emerges is smaller, more local, and slower to centralize: the dark age, and the parochial world that follows it—the manorial village, the walled town, the local lord who answers to no one. The Bronze Age collapse, the centuries after Rome, and the more recent cascades of finance and supply all share this shape: a web that was too dense to fail safely, and a long, quiet, local recovery.

The World Right Now

This is not only a theory of antiquity. Applied to the present, it names three structural facts.

First: the great modern asymmetry is not what people earn. It is who can leave. Capital can exit almost costlessly—a wire, a flag of convenience, an offshore address. Labour cannot: it has visas, mortgages, language, children, and borders. The result is a structural transfer that needs no conspiracy, only asymmetry. Whenever a society debates inequality, the theory says the deeper question is not *what share does each side get* but *how cheaply can each side walk away*.

Second: for the first time, the host has no outside. Every predator in history could move on: new lands, new peoples, new rivers. Carbon extraction is predation on an atmosphere with no replacement and no escape—and the theory's law of patient predation says that a patient predator conserves its host. The most consequential number in the modern ledger is therefore not any tax rate or wage. It is civilisation's **discount rate**: how much we value the future against the present. Every other variable follows it.

Third: an order can now decay without falling. For most of history, lost belief meant rebellion, and rebellion meant collapse. Modern orders have quietly substituted architecture for belief: surveillance that makes dissent expensive, narratives so fragmented that grievance never gathers, and just enough welfare to prevent outright immiseration. The result is a strange new state of things—**stable, and not legitimate**—in which the ledger tilts further every year and nothing appears to happen. The theory's warning is that this stability is thin. It holds until a single shock—a fiscal crisis, a technology that defeats surveillance, a story that unifies what was fragmented—and then the translation of years of grievance into conflict can happen in weeks.

The hidden history of the present is already visible in the theory's method. Just as the pyramids concealed the corvée, the smartphone conceals the cobalt miner, the algorithm conceals the data annotator, and the supply chain conceals the garment worker. The achievement is recorded. The labour is not. Measure the ledger first, and the record rearranges itself.

What to Ask, What to Hope For

The theory does not tell you what to admire. It tells you what to measure. Wherever you see a civilisation—your own country, your company, a marriage, an alliance, an empire—ask five questions:

1. **Who gives?**

2. **Who gains?**
3. **Who cannot leave?**
4. **Who cannot speak?**
5. **Who is missing from the record entirely?**

The fifth question is the newest, and the sharpest. Every unbalanced order hides some of its costs outside its own accounting: the unpaid work that never appears in the product, the pollution settled on people who were never asked, the future generations who will pay for this decade's excess and will never cast a vote. An order that cannot see the people it is quietly draining will keep draining them, sincerely, right up to the arithmetic that destroys it.

What to hope for, then, is specific and not utopian. Not a world without ledgers—there is no such thing. A world in which the ledgers are **close enough to balance that no one is consumed**, in which people can leave what harms them and speak about what is done to them, in which the machinery of rebalancing runs faster than the drift, and in which the powerful have reason—not just fear—to keep the host alive.

That is what the word "civilisation" was always supposed to mean. The monuments are optional.

Who gives, who gains, who cannot leave, who cannot speak, who is missing from the record entirely.